ANSWER KEY & MARKING SCHEME SET 1

Chapter 8: **Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** vibration	Vibration.
2	**(d)** above 20 kHz	Higher.
3	**(b)** 100 oscillations per second	Definition.
4	**(d)** medium	To travel.
5	**(a)** below 20 Hz	Lower.
6	**(c)** 0.1 s	1/f.
7	**(c)** vacuum	No medium.
8	**(c)** reflection of sound	Reflection.
9	**(d)** wavelength	v = f lambda.
10	**(a)** longitudinal wave	Compressions.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	A form of energy produced by vibrating bodies (2).	2
12	Distance between two consecutive compressions or crests (2).	2
13	SONAR (1); Medical imaging (1).	1+1
14	A disturbance that carries energy through a medium (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Tuning fork (1); Drum (1); Vocal cords (1).	1+1+1
16	T = 1/f (1); 1/250 (1); 0.004 s (1).	1+1+1
17	Increases with temperature (1); Warm air (1); Cold air slower (1).	1+1+1
	OR: Direction of vibration (1); Compressions vs crests (1); Sound vs rope waves (1).	1+1+1
18	SONAR (1); Bats finding prey (1); Depth of sea (1).	1+1+1
	OR: Infrasonic below 20 Hz (1); Ultrasonic above 20 kHz (1); Elephants, bats (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Vibration (1); Medium (1); Compression (1); Rarefaction (1); Wave travels (1).	1+1+1+1+1
	OR: Audible (1); Infrasonic (1); Ultrasonic (1); Examples (1); Uses (1).	1+1+1+1+1
20	Crest (1); Trough (1); Amplitude (1); Wavelength (1); Frequency (1).	1+1+1+1+1
	OR: Emits ultrasound (1); Reflect (1); Detects echo (1); Finds prey (1); Avoids obstacles (1).	1+1+1+1+1

Mistakes to avoid:

- Forgetting to halve the total distance in echo and SONAR problems: the sound goes to the obstacle *and back*, so $d = vt/2$.
- Mixing up pitch and loudness: pitch depends on frequency, loudness on amplitude.
- Drawing sound as a transverse wave or saying that the particles travel with the wave; in air the particles vibrate to and fro along the direction of travel.
- Saying that sound travels fastest in air and slowest in solids; the order is solids $>$ liquids $>$ gases.
- Writing the audible range as 20 kHz to 20 Hz, or calling a sound of 25 kHz infrasonic.
- Applying $v = f\lambda$ with wavelength in cm or frequency in kHz without converting to m and Hz, or saying that the speed of a wave changes when only the frequency is changed in the same medium.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 8: **Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 0.1 s	Persistence.
2	**(b)** 100 m/s	f lambda.
3	**(b)** hertz	Hz.
4	**(c)** ultrasound	Echolocation.
5	**(d)** 2 m	v/f.
6	**(b)** frequency	Frequency.
7	**(d)** ultrasonic waves	Echo.
8	**(c)** 340 Hz	v/lambda.
9	**(a)** amplitude	Amplitude.
10	**(a)** light is faster than sound	Speeds differ.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Reflected sound heard after the original (2).	2
12	20 Hz to 20000 Hz (2).	2
13	Number of oscillations per second (2).	2
14	Sensation of the frequency of sound (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Pitch (1); Loudness (1); Quality (1).	1+1+1
16	Medical scan (1); Cleaning (1); Detecting cracks (1).	1+1+1
17	Metal of bell (1); Strings of guitar (1); Air in a flute; membrane of drum (1).	1+1+1
	OR: v = f lambda (1); 100 x 3 (1); 300 m/s (1).	1+1+1
18	Plant trees (1); Silencers (1); Limit loud speakers (1).	1+1+1
	OR: Energy (1); Information (1); Not matter (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Pitch (1); Loudness (1); Quality (1); Dependent on (1); Examples (1).	1+1+1+1+1
	OR: Meaning (1); Cause (1); Effect (1); Curtains (1); Carpets (1).	1+1+1+1+1
20	Definition (1); Reflection (1); 0.1 s (1); 17 m (1); Example (1).	1+1+1+1+1
	OR: Tuning fork (1); Pith ball (1); Strike (1); Ball moves (1); Vibration (1).	1+1+1+1+1

Mistakes to avoid:

- Forgetting to halve the total distance in echo and SONAR problems: the sound goes to the obstacle *and back*, so $d = vt/2$.
- Mixing up pitch and loudness: pitch depends on frequency, loudness on amplitude.
- Drawing sound as a transverse wave or saying that the particles travel with the wave; in air the particles vibrate to and fro along the direction of travel.
- Saying that sound travels fastest in air and slowest in solids; the order is solids $>$ liquids $>$ gases.
- Writing the audible range as 20 kHz to 20 Hz, or calling a sound of 25 kHz infrasonic.
- Applying $v = f\lambda$ with wavelength in cm or frequency in kHz without converting to m and Hz, or saying that the speed of a wave changes when only the frequency is changed in the same medium.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 8: **Sound Waves: Characteristics and Applications** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 1 s	340/340.
2	**(d)** solids	Particles close.
3	**(c)** sound of a single frequency	Pure note.
4	**(d)** 680 m	vt.
5	**(a)** metre	m.
6	**(c)** multiple reflections	Sound guided.
7	**(d)** 3 s	1000/340.
8	**(d)** 20 Hz to 20 kHz	Human.
9	**(b)** curtains	Absorbers.
10	**(a)** high pressure	Crowded.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	No medium (1); Needs particles (1).	1+1
12	Maximum displacement from the mean position (2).	2
13	Sensation related to the amplitude of sound (2).	2
14	v = f lambda (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	lambda = v/f (1); 330/110 (1); 3 m (1).	1+1+1
16	Light faster (1); Sound slower (1); Gap increases with distance (1).	1+1+1
17	d = vt (1); 340 x 3 (1); 1020 m (1).	1+1+1
	OR: f = v/lambda (1); 340/0.5 (1); 680 Hz (1).	1+1+1
18	Air 340 m/s (1); Water 1500 m/s (1); Steel 5000 m/s (1).	1+1+1
	OR: Distance too small (1); Time gap less than 0.1 s (1); Overlaps with the original sound (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Ultrasonic pulse (1); Reflection (1); Time measured (1); d = vt/2 (1); Uses (1).	1+1+1+1+1
	OR: Solids (1); Liquids (1); Gases (1); Reason (1); Temperature (1).	1+1+1+1+1
20	Total distance = 680 m (1); Distance = 340 m (1); d = vt/2 (1); Answer (1); Units (1).	1+1+1+1+1
	OR: Meaning (1); Sources (1); Effects (1); Control (1); Law (1).	1+1+1+1+1

Mistakes to avoid:

- Forgetting to halve the total distance in echo and SONAR problems: the sound goes to the obstacle *and back*, so $d = vt/2$.
- Mixing up pitch and loudness: pitch depends on frequency, loudness on amplitude.
- Drawing sound as a transverse wave or saying that the particles travel with the wave; in air the particles vibrate to and fro along the direction of travel.
- Saying that sound travels fastest in air and slowest in solids; the order is solids $>$ liquids $>$ gases.
- Writing the audible range as 20 kHz to 20 Hz, or calling a sound of 25 kHz infrasonic.
- Applying $v = f\lambda$ with wavelength in cm or frequency in kHz without converting to m and Hz, or saying that the speed of a wave changes when only the frequency is changed in the same medium.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.